

STATUS of the HALL A MØLLER POLARIMETER

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MØLLER SYSTEMATIC ERROR BUDGET

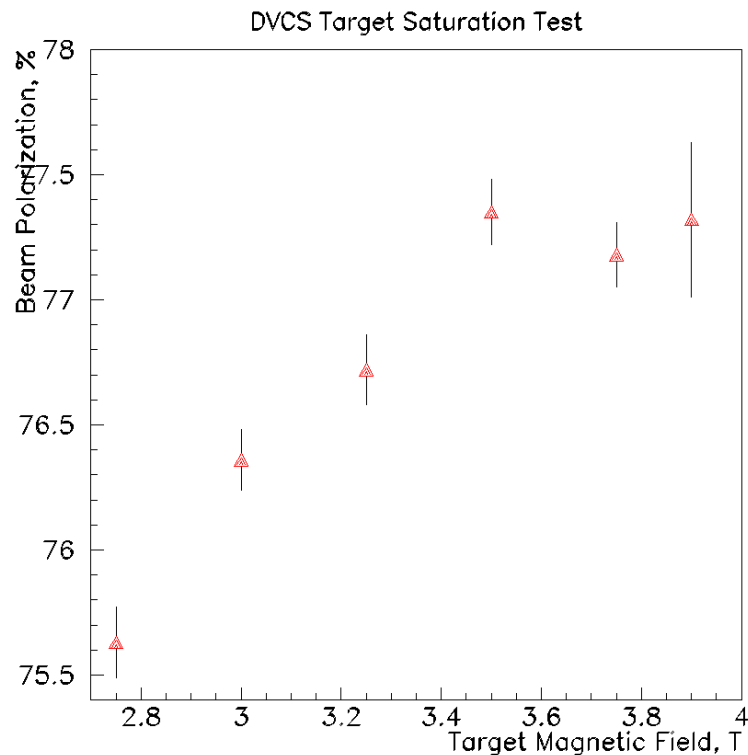
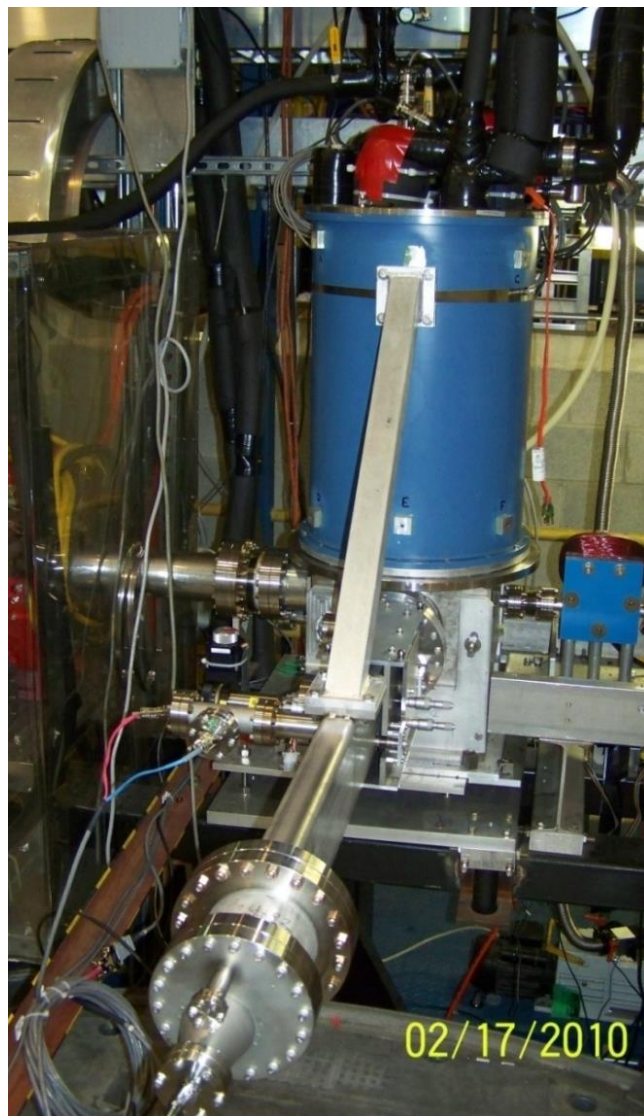
Variable	2010 (3.7T)	PREX-II
<i>Target polarization</i>	0.35%	0.25%
<i>Analyzing power</i>	0.3%	0.2%
<i>Levchuk effect</i>	0.3%	0.2%
<i>Target temperature</i>	0.02%	0.02%
<i>Dead time</i>	0.3%	0.05%
<i>Background</i>	0.3%	0.2%
<i>High beam current*</i>	0.2%*	0.2%
<i>Others</i>	0.5%	0.3%
Total	0.87%	0.53%

* M. Poelker et al. Phys.Rev.ST Accel.Beams 10 (2007) 053502

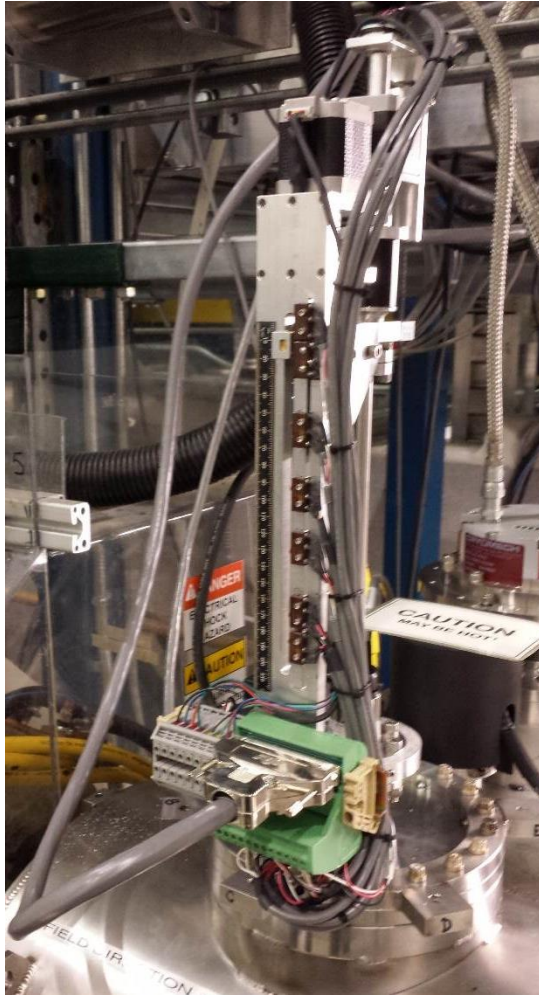
HALL A "BRUTE FORCE" TARGET 2010

Problems:

SC magnet required liquid N and low pressure liquid He
complex and expensive operation
daily refill with liquid He
4T max Field, >3.5T for the target saturation
No remote target alignment
No success with magnet field alignment along the beam



NEW POLARIZED ELECTRON TARGET



Hall A High-Field Moller Target Motion Control

Linear

Motor Readback (mm) **0.000** Encoder (v) **3.031** Reset

Position (mm)	0	43.444	80.265	117.483	155.298
Encoder (v)	3.025	2.364	1.813	1.251	0.680

Retracted Park T1 T2 T3 T4 Extended

P T1 T2 T3 T4 Move

0.000 < **0.000** > STOP

Position Input (mm) Jog (mm)

SM

Rotary

Motor Readback (degree) **-0.112** Encoder (v) **4.568**

Position (degree)	-8.426	-0.200	12.865
Encoder (v)	4.150	4.558	5.206

-Limit C +Limit

-L C +L Move

-0.112 < **0.000** > STOP

Position Input (degree) Jog (degree)

SM



Designed and made by Temple University

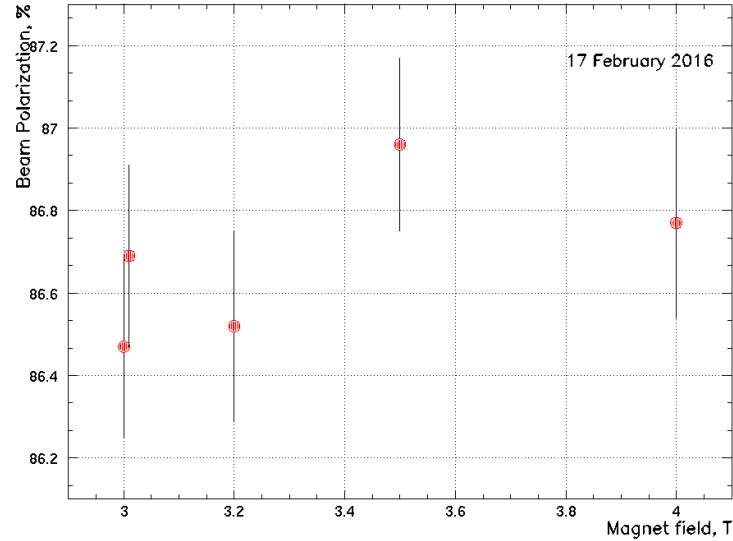
Rotation in horizontal plane $\pm 10^\circ$

Vertical translation

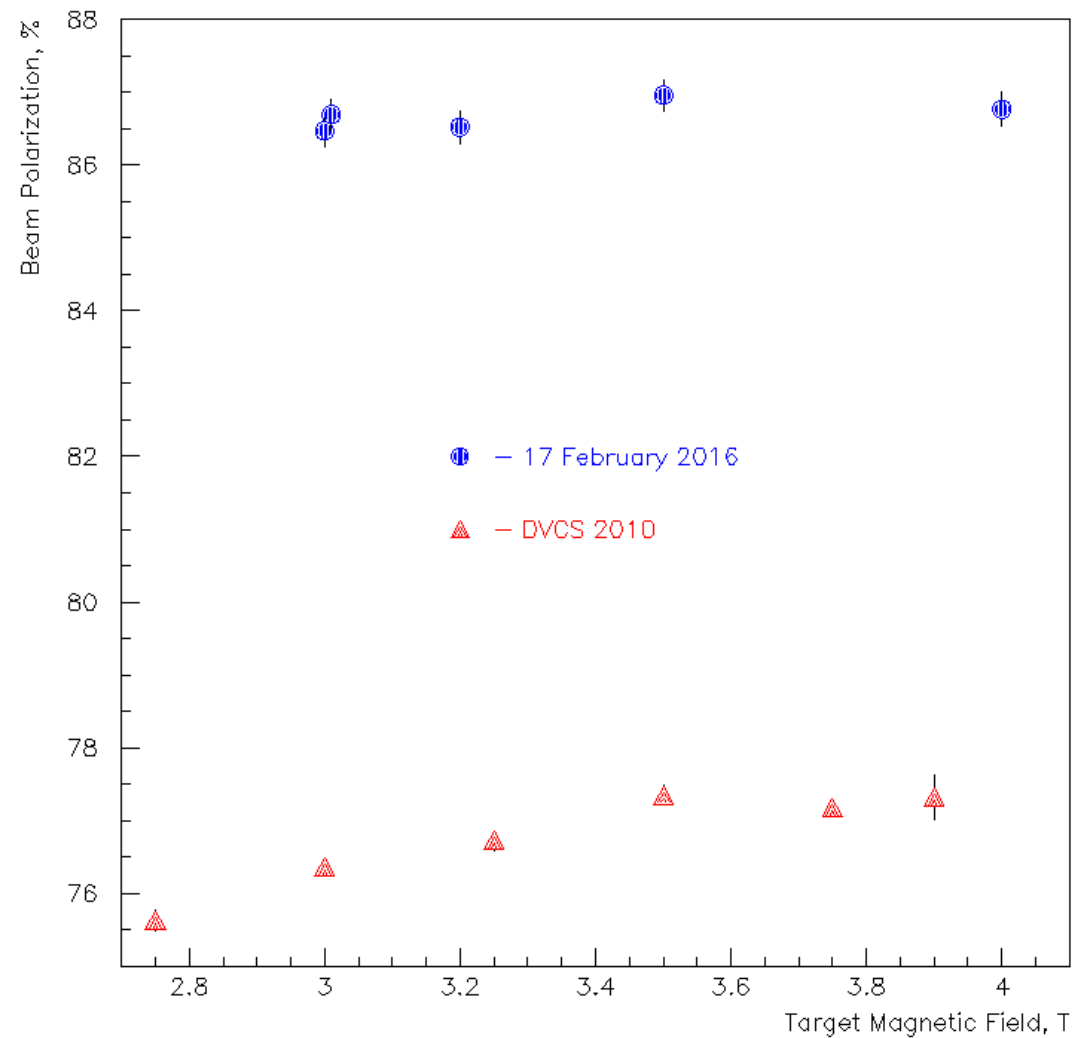
Targets in holder: pure iron $1\mu\text{m}$, $4\mu\text{m}$, $12\mu\text{m}$, $25\mu\text{m}$

NEW TARGET SATURATION TEST

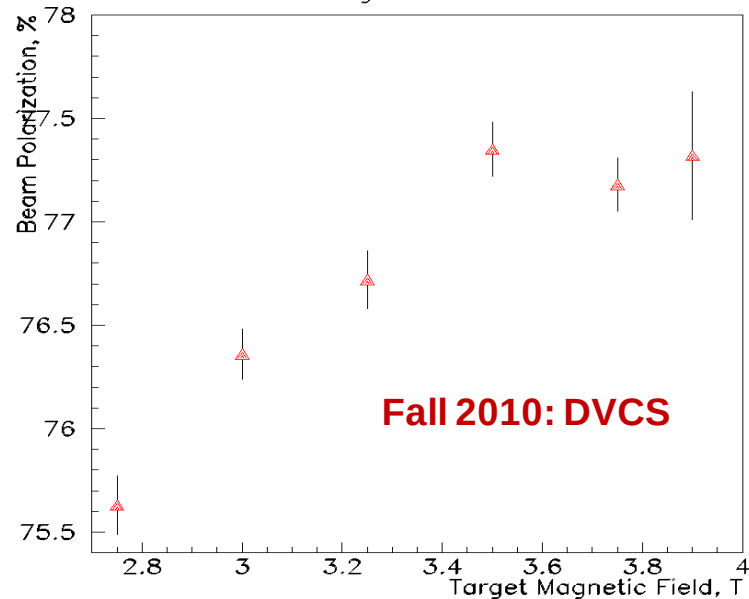
Measured Polarization vs. Target Saturation Field
Moller Polarimeter



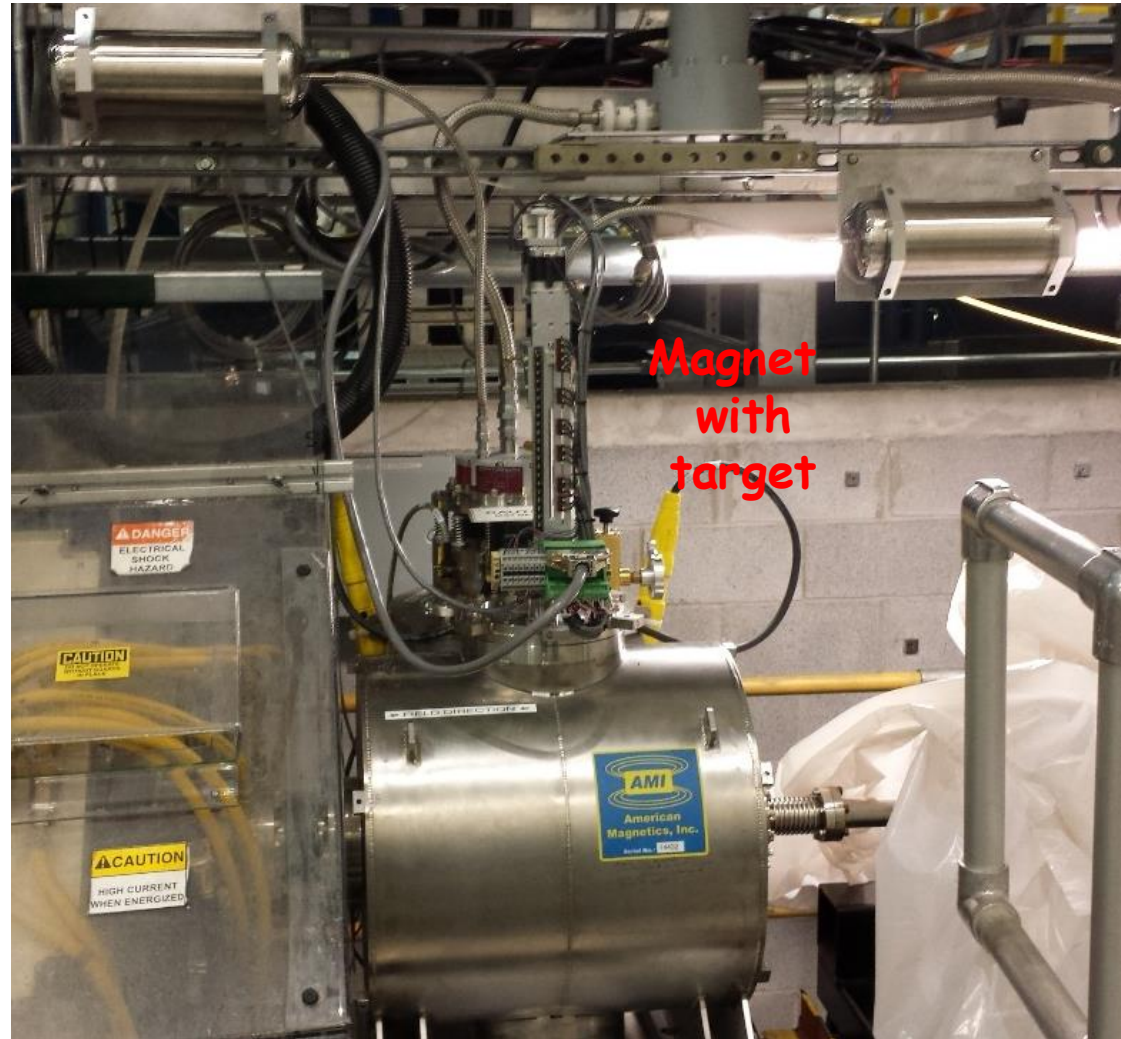
DVCS Target Saturation Test



DVCS Target Saturation Test



NEW MAGNET



New superconductive magnet:

Maximal field $\pm 5T$

Very limited information

- portable*
- no LN required*
- no cryo lines required*

Small amount of He →

- low thermal capacity and “cold” production →*
- slow field ramp-rate ($\sim 35\text{min: } 0 \rightarrow 3T$)*

EXPERIENCE WITH MAGNET OPERATION

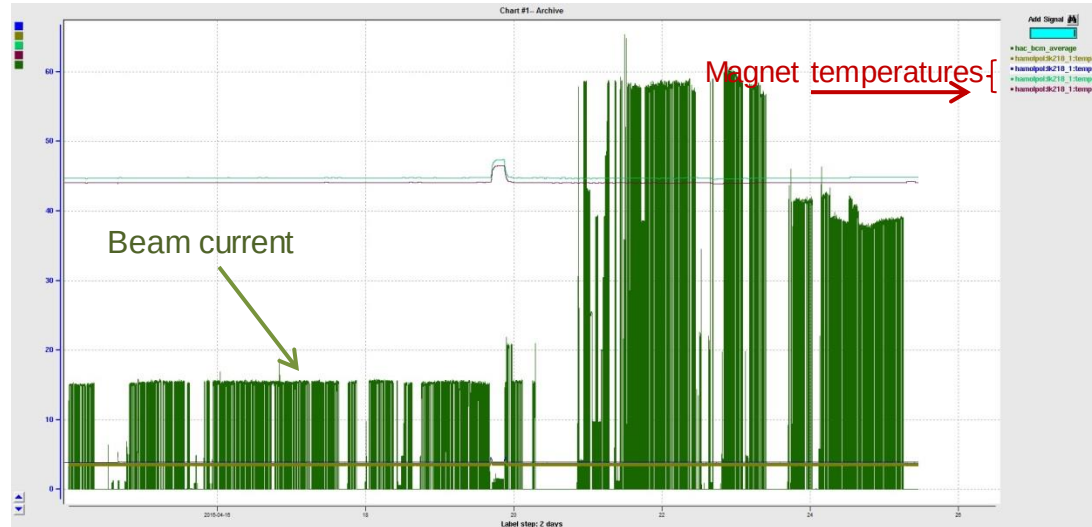
Magnet can be used at high beam current and high radiation in Hall A

04/19/2016 – First Moller measurement with beam through the Compton chicane (11GeV)

High radiation:

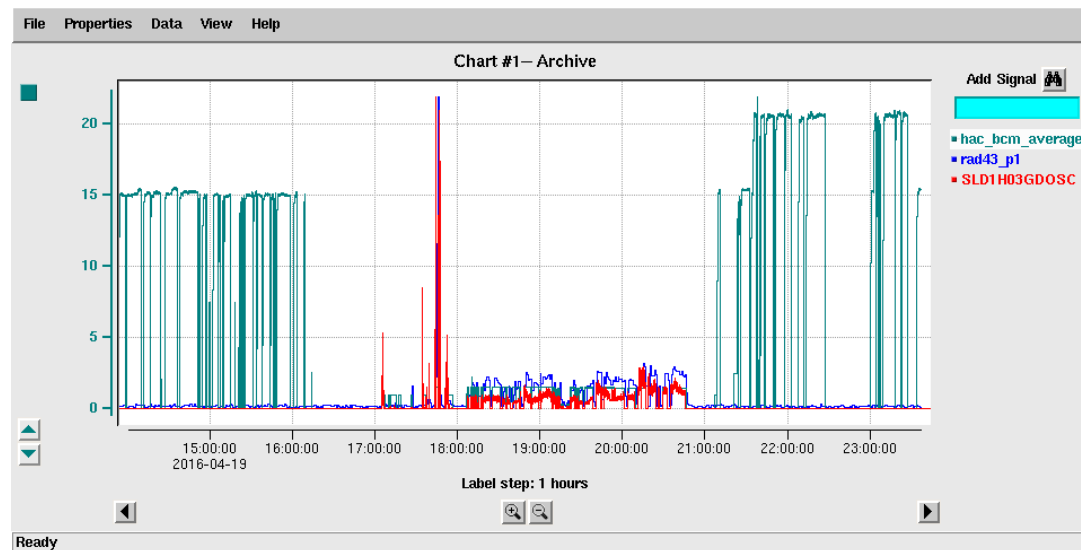
- *Hall A target in “EMPTY” not at “HOME”;*
 - *Optics combined with Compton;*
 - *Combination of all above.*
- Will be checked in the fall run.*

Magnet alignment test at low beam energy is needed.

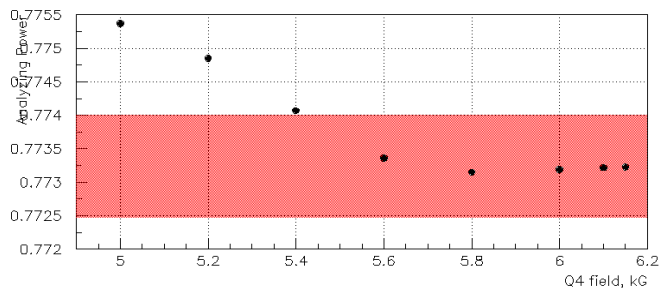
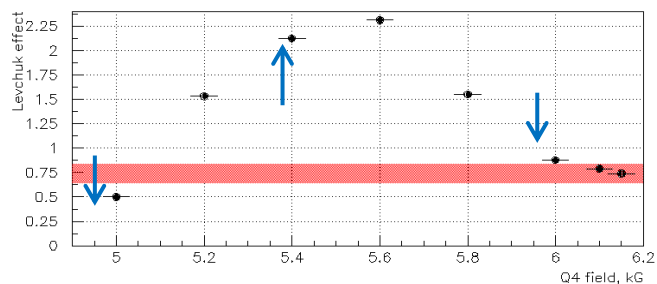
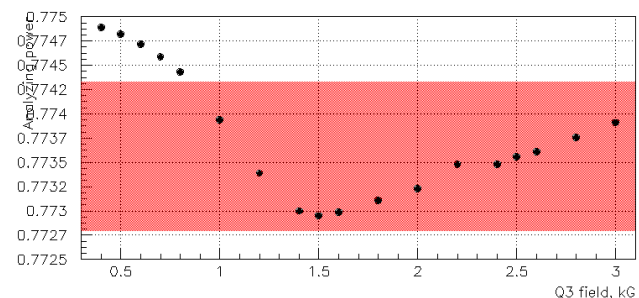
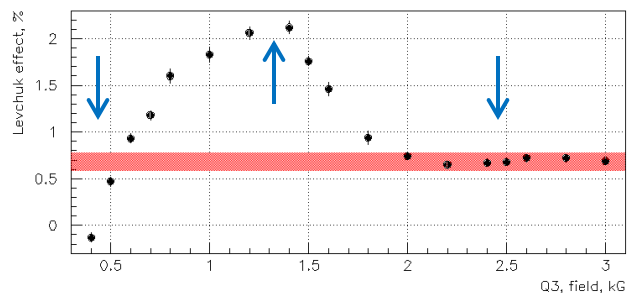


Rad43_p1 – Hall A track ramp (up to 0.9 mrem/h)

SLD1H03GDOSC – target ion chamber (up to 50 rad/h)

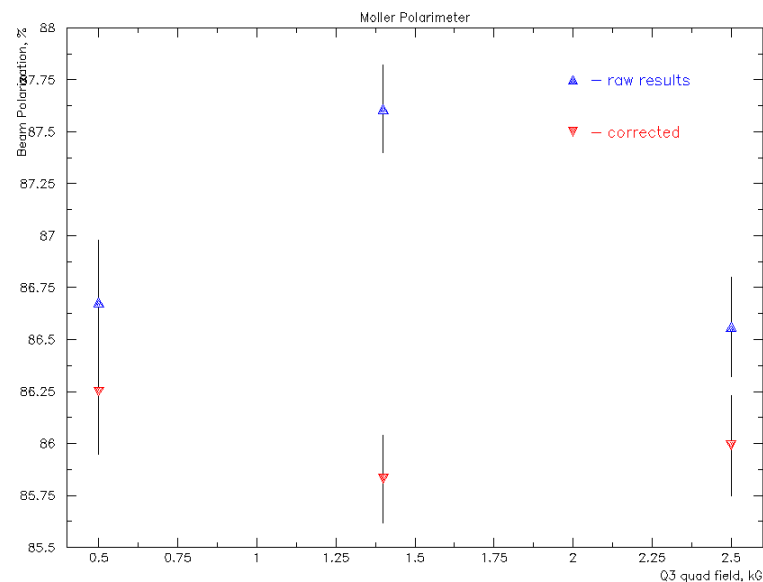


GEANT vs. MEASUREMENT SYSTEMATICS

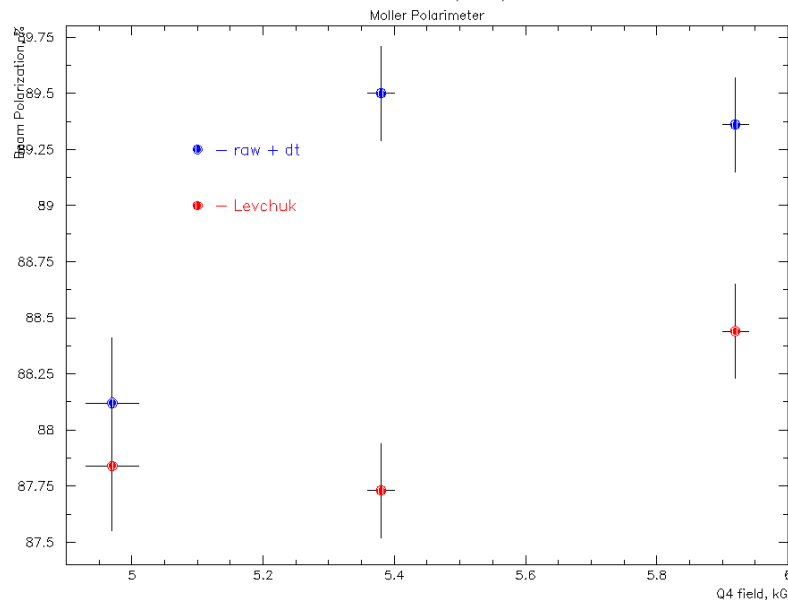


$E_b = 11 \text{ GeV}$

Moller results, 03/31/2016



Moller results, 04/19/2016



MORE SYSTEMATICS

Dead time:

- measure it with using LED pulser***

More problems:

- Background, rad. corrections;***
- Fast beam position feed-back for the Møller target?***
- Dithering vs. SC magnet turned on?***



KERR EFFECT

The Kerr magneto-optic effect:

Was successfully used off-line in Hall C long time ago;

Unique facility to test the target mechanism/alignment with no beam

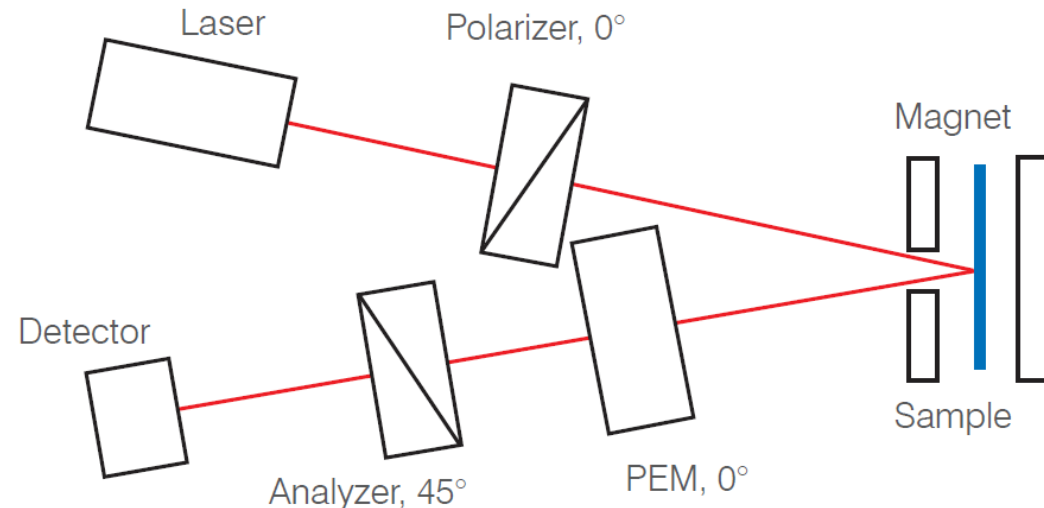
Polar Kerr effect is in one magnitude larger than Longitudinal or Transverse Kerr effect.

The magnetization is perpendicular to the surface, parallel to the direction of the incident linearly polarized light: the polarization is rotated through an angle α_K given by

$$\alpha_K = K_R M$$

K_R - is the Polar Kerr constant

M – is the target magnetization



CONCLUSION

Magnet: *more experience with operation - fall 2016*

Target: *upgrade - summer 2016*

commissioning: on-line - fall 2016

off-line (Kerr) - winter-summer 2017

Spectrometer + DAQ:

GEANT simulation

GEANT vs measurements - fall 2016

